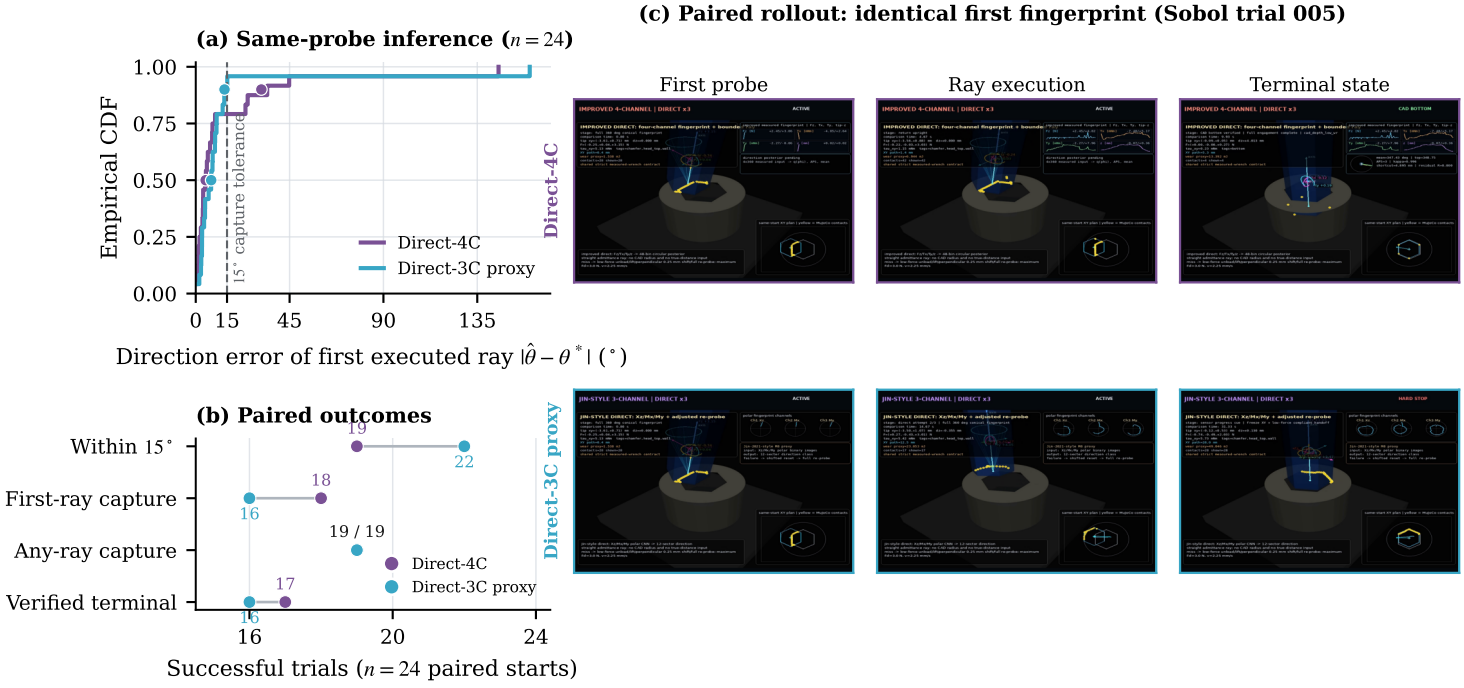

(a) Same-probe inference ($n = 24$)
Empirical CDF
1.00
0.75
0.50
0.25
0.00
0
15
45
90
135
15° capture tolerance
Direct-4C
Direct-3C proxy
Direction error of first executed ray $|\hat{\theta} - \theta^*|$ (°)
(b) Paired outcomes
Within 15°
First-ray capture
Any-ray capture
Verified terminal
19
22
18
16
19 / 19
17
16
16
20
24
Direct-4C
Direct-3C proxy
Successful trials ($n = 24$ paired starts)
(c) Paired rollout: identical first fingerprint (Sobol trial 005)
First probe
Ray execution
Terminal state
Direct-4C
Direct-3C proxy